Activity #1: Initial Learning

Listed below is a variety of initial resources that you can use if you are not familiar with electrical engineering concepts. I would suggest getting to know these well before you attempt to work on the buck converter. Any resource that talks about the same thing is good, as long as you understand the basics.

Analog Devices and Circuits:

1. Organic Chemistry Tutor: Ohm's Law
   https://www.youtube.com/watch?v=_rSHqvjDksg
2. Khan Academy, Current, Resistors, Ohm's Law:
   https://www.khanacademy.org/science/ap-physics-2/x0e2f5a2c:electric-circuits/x0e2f5a2c:current-resistivity-ohms-law/v/circuits-part-1
3. Khan Academy, Compound Circuits:
   https://www.khanacademy.org/science/ap-physics-2/x0e2f5a2c:electric-circuits/x0e2f5a2c:current-resistivity-ohms-law/v/electric-power
4. EEVBlog, Voltage Dividers:
   https://www.youtube.com/watch?v=xSRe_4TQbuo&list=PLvOlSehNtuHtWlH0UOZNtOn-FlFCn1GYw&index=2
5. EEVBlog, Kirchhoff's Laws:
   https://www.youtube.com/watch?v=WBfAEeEzDlg&list=PLvOlSehNtuHtWlH0UOZNtOn-FlFCn1GYw&index=13

Listed below are more power related resources. This you will need to use to understand the theory behind how switch mode power supplies work.

1. MIT OpenCourseWave Power Electronics Course:
   a. Lecture 5: Intro to DC/DC, Part 1
2. Fundamentals of Power Electronics, by Robert W. Erickson and Dragan Maksimovic
3. Power Electronics by Daniel W. Hart

Please take good notes! Take notes in your notebook, here, whatever you would want to do. I would just like to see what you guys see and how you understand it!

Notes Below

MIT Lec 1 Key takeaways:

- Voltage dividers are inefficient

- We can use switches to achieve an average voltage matching our desired output V across load
    - But, must include
- Ideal switches are lossless
- Buck = step-down, boost = step-up, buck-boost = either

Hierarchy
- Voltage Regulator
    - Linear regulator
        - Highly inefficient but cheaper & simpler
    - Switching regulator
        - Buck IC
            - Contains the control logic, feedback loop, and FETs
            - Does NOT contain the LC filter, which completes the buck converter circuit
        - Boost IC
        - Buck-boost IC

Physical Notes Next Page

MIT Opencourseware – Power Electronics: Intro to DC/DC Part 1

Start w/ KCL:

$$\sum_{n=1}^{k} i_n = 0$$

average it

$$\left\langle \sum_{n=1}^{k} i_n \right\rangle = 0$$

$$\boxed{\sum_{n=1}^{k} \langle i_n \rangle = 0}$$

same for KVL:

$$\boxed{\sum_{n=1}^{k} \langle v_n \rangle = 0}$$

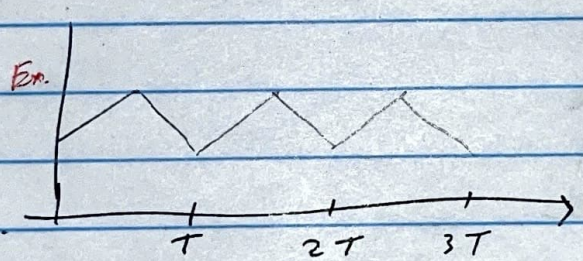
New analysis techniques
- AVERAGE KCL &
- AVERAGE KVL

idk what PSS is...

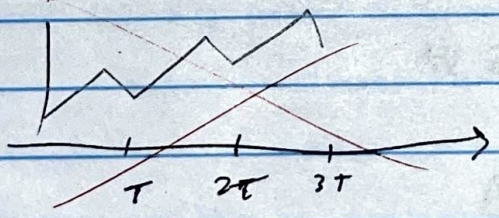
going back to Lecture 2: Analysis Methods and Rectifiers

Periodic Steady State (PSS)

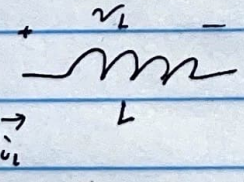
Ex.



repeated waveform over a T



NOT PSS



- $v_L = L \frac{di}{dt}$

look at avg

$\langle v_L \rangle = \langle L \frac{di}{dt} \rangle$

$\langle v_L \rangle = L \langle \frac{di}{dt} \rangle \rightarrow$ in PSS, $= 0 \Rightarrow \langle v_L \rangle = 0$ in PSS

PERIODIC STEADY STATE conditions

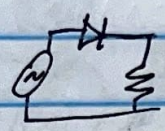
so:

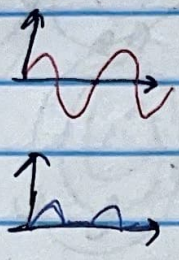
Inductors $\langle V_L \rangle = 0$ in PSS

Capacitors $\langle I_C \rangle = 0$ in PSS

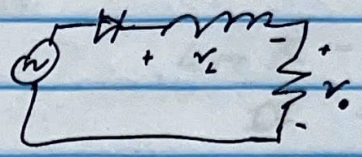
↳ extension: @ DC, L → short
C → open

Simplest Rectifier





Add an Inductor

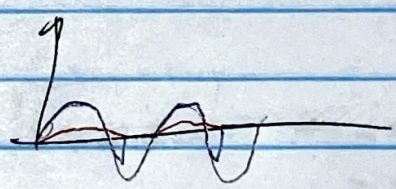


KVL: $\langle V_s \sin(\omega t)\rangle - \langle V_L\rangle - \langle V_o\rangle = 0$

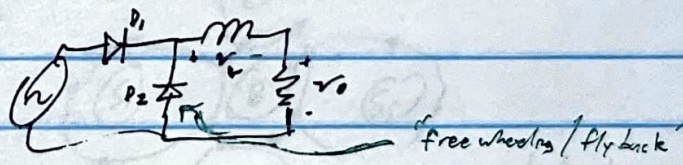
$0 \quad - 0 \text{ (across)} - 0 = 0$

- when $V_s \sin(\omega t)$ goes negative, L continues pushing current, keeping diode on, letting it get some of the negative supply voltage until it evens out the pull of the L, then diode shuts off since $i = 0$
  - can't go negative (diode not allow)

(V_L / back EMF)
- changing w/ $\frac{di}{dt}$



Add 2nd Diode



- now inductor current has a path to continue, no longer fighting V_{supply}
- i_L can fade more gradually

- when V_s goes negative D_1 off, D_2 on

$v_L = L\frac{di}{dt}$

$i_c = C\frac{dv_c}{dt}$

MIT

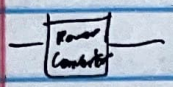
↳ going back to PE lecture 5

$q(t)$: (switching function)

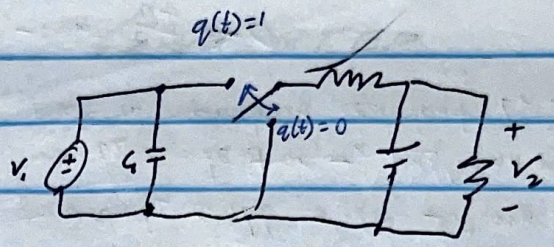
IN PSS

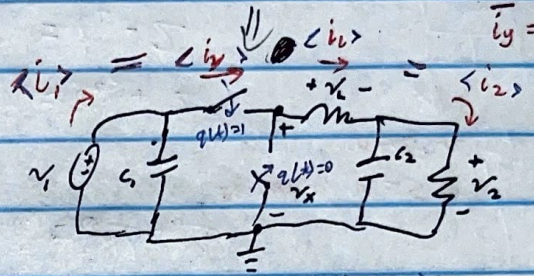
i_c: $\frac{1}{T}\int_0^T i_c = \langle i_c \rangle = 0$

v_L: $\frac{1}{T}\int_0^T v_L = \langle v_L \rangle = 0$



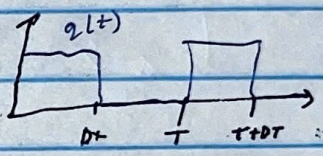
if $\eta = 1$, $P_{in} = P_{out}$



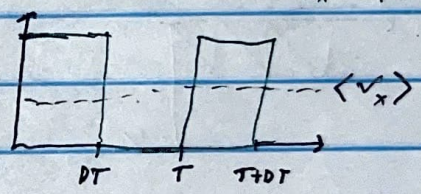


$\bar{i_y} = \bar{i_1} = \bar{i_2}$

• no $\langle i_c \rangle$ in PSS / = 0



• $v_x = v_1 q(t)$



Ideal Buck Conv. in PSS

"output" = V depends on "input" V_{in}

$V_{out} = DV_{in}$

$I_{in} = DI_{out}$ OR $I_{out} = \frac{1}{D} I_{in}$

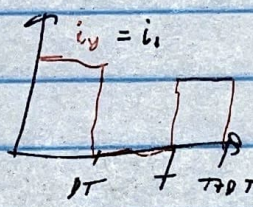
($I_{out} = I_L$)

"output" I_o depends on "load"

↳ $P_{in} = I_{in} V_{in} = I_{out} V_{out} = P_{out}$

Power conservation → DC/avg input & output relationships in PSS

$I_2 = I_L$



$I_y = DI_L = I_1$

$I_1 = DI_2$

KVL: (for both on $q(t)$)

$\langle v_x \rangle - \langle v_L \rangle - \langle v_2 \rangle = 0$

$\Rightarrow \langle v_2 \rangle = \langle v_x \rangle$

$V_2 = DV_1$

↳ V_o depends on Duty Ratio

Big Picture Idea:
- AVERAGE Values
- DC, steady-state

DC component? / DC value? = AVG

- the 0 Hz / zero frequency term
- think Fourier analysis gets the constant term A_0 or $\frac{a_0}{2}$ or a_0

$f(x) = A_0 + \sum_{n=1}^{\infty} A_n \cos(n\omega t + \theta_n)$

Look! it's defined as the AVERAGE of the period

Where $A_0 = \frac{1}{T}\int_0^T f(t)\,dt$

• if $n=0$, you get the 0-Hz DC comp. do only

Huge L and Huge C Idealizations

$v_L = L\frac{di}{dt}$

$\frac{di}{dt} = \frac{v_L}{L}$ $\xrightarrow{L\to\infty}$ $\frac{di}{dt} \to 0$

so $i_L \approx$ constant during switching cycle

w/ large L_s and C_s in buck converter,

i.e. neglecting the fluctuating waveforms w/ its average

$i_L(t) \approx I_L$

$v_c(t) \approx V_c$

$i_c = C\frac{dv_c}{dt}$

$\frac{dv_c}{dt} = \frac{i_c}{C}$ $\xrightarrow{C\to\infty}$ $\frac{dv}{dt} \to 0$

so $V_c \approx$ constant during switching

Activity #2: Component Selection

After finishing Activity #1, you are ready to start selecting different components! When selecting components, please follow the LV Standards guideline to picking different form factors. Typically, our resistors and capacitors are both 0603, but this is just a general guideline. Typically do not go under 0603, but if we need a larger voltage rating, going up is completely fine. We typically only source from Digikey as well (Mouser if it is the only place available). We also do not want to use marketplace products. These are all filters that you can apply on Digikey, and more information is available in the LV standards document.



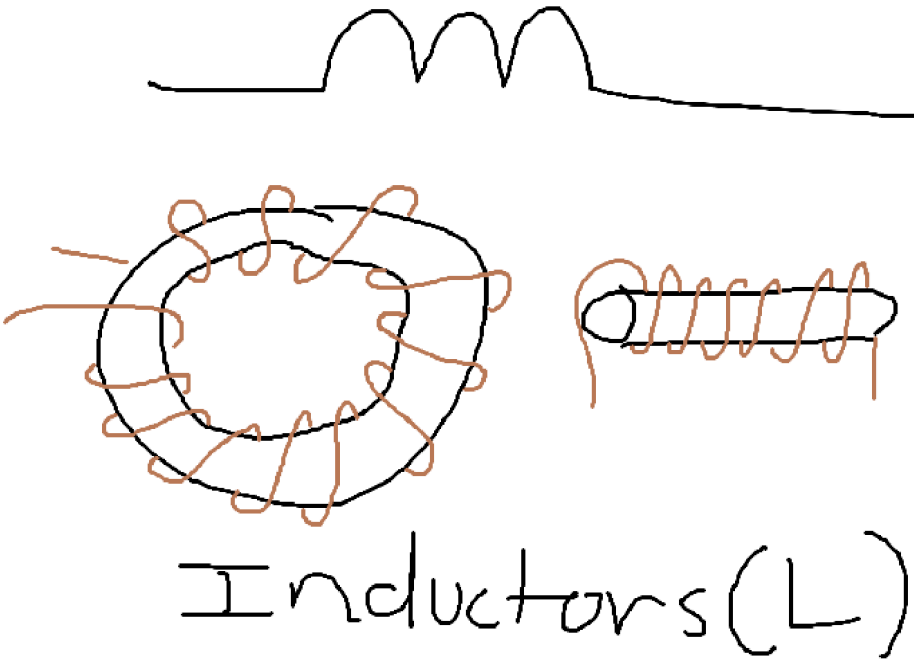
The first big thing to source is your Buck IC! This chip is used to adjust the duty cycle of the switches, and work with the feedback loop. For reference, here is the one we used last year but is now out of stock. The big thing is making sure it can output enough current to supply our boards! For FS5, we defaulted to a 2A buck, which worked perfectly fine (to my knowledge), so anything 2A and above is good! However, be sure to properly select this, because there are some tradeoffs that you may have to make if you oversize the buck converter too much. Feel free to ask me if you have any questions about that!

There are a couple I have in mind, but I would like you guys to do your own research and play with Digikey to be more familiar with the website! This is where we order almost all our parts, excluding some from Mouser. You will need to get familiar with it. Here are some initial parameters to get you all started on looking.

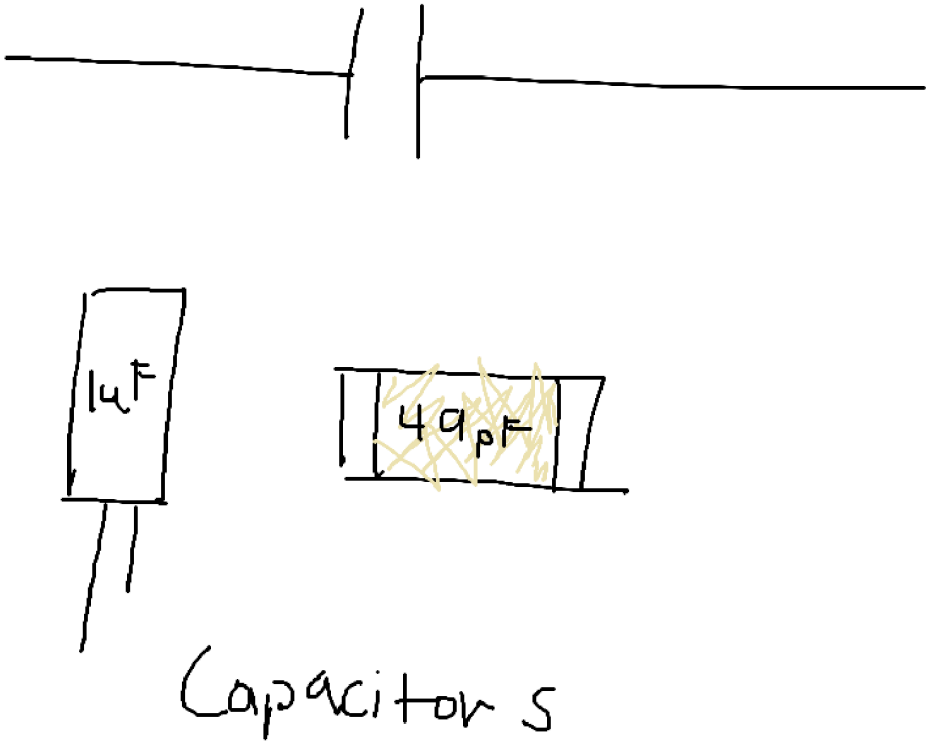
Input Voltage	~10-15V *
Output Voltage	5V
Output Current	Iout ≥ 2A
Switching Frequency	Fsw ≥ 500kHz**
Feedback Control	😳
Ripple Voltage	1-5%

*Though our LV bus is 12V nominally, it is actually 13.5V!
**with too low of a switching frequency, you upsize your inductor and output capacitor value, which could be an issue space wise. This is an arbitrary value I have set for the case of the onboarding project.

Whatever chip you decide to pick, be sure to properly read the entire thing, understanding what each pin does, what kind of internal structure it has, and the reasoning behind all of it.



One big thing you should check in an inductor is its saturation current. Make sure that your inductor does not oversaturate at the output current you are expecting, or else your inductor's inductance will drop so low that your output ripple will be extremely large.



Capacitors have a funny little thing called DC Biasing. Capacitors do not like staying at the specific capacitance they were designed to be at. Instead, whenever you apply any kind of voltage across one, their capacitance will drop by some percentage, typically shown in your datasheet. Make sure that your capacitances will be what you are expecting at your expected output voltage! A little hint: The output capacitors are there to help filter noise! There is something you have to do when picking your output capacitors to help reduce noise on the output end.



Make sure to try to minimize costs as much as possible! This is more of a good engineering practice to not just go with the best thing if it costs an unreasonable amount of money.

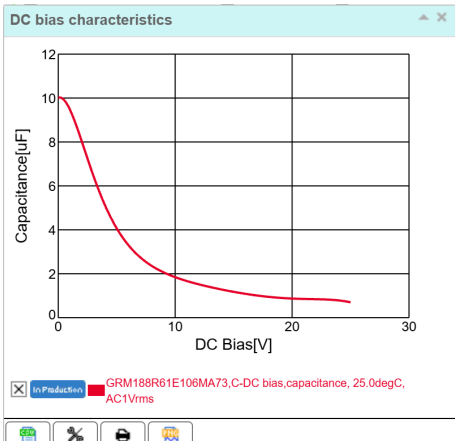
At the end of this activity, fill out this chart with the things that you have selected for ONE buck converter, with the costs per unit and size as well. If you need to add more rows, feel free to.

Component	Cost	Link	Units	Size	Value(s)
Buck IC	1: $0.27000 Cut Tape (CT)	TPP302080-T6TR	1x		V_in = 4.5V–30V V_out = 0.6V–30V I_out = 2A f_s = 500kHz
Inductor	1: $0.37000 Cut Tape (CT)	HCM1A0805V3-150-R - Low on stock, 2nd option in case: VLS6045EX-150M	1x	8.10mm x 8.00mm x 5.40mm	L = 15µH I_sat = 4.6A DCR = 52mΩ
Capacitor (output)	~~1: $0.24000~~ ~~Cut Tape (CT)~~ 1: $0.37000 Cut Tape (CT)	~~GMC10X5R226M10NT~~ GRM187R61A226ME15D	2x	0603	22µF 10V
Capacitor (input)	1: $0.28000 Cut Tape (CT)	GRM188R61E106MA73J	2x	0603	10µF 25V
Capacitor (decoupling & bootstrap)	1: $0.10000 Cut Tape (CT)	CGA3E2X7R1E104K080DA	3x	0603	100nF 25V
Resistor	1: $0.10000	RG1608P-22	1x	0603	22kΩ

(feedback)	Cut Tape (CT)	3-B-T5			
Resistor (feedback)	1: $0.11000 Cut Tape (CT)	RG1608P-302-B-T5	1x	0603	3kΩ
Total Cost	**$2.45000**				

Selection Notes

GRM188R61E106MA73J (10µF input caps)



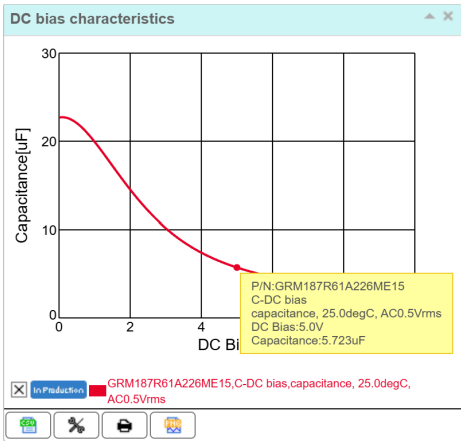
Wow, looks bad!
- Only 1-1.5µF at 10-15V input,
- Will produce ~0.30V pp input ripple using this equation (worst case D=50%):

$$\Delta V_{IN(C)} \approx \frac{I_{out} \cdot D \cdot (1 - D)}{f_{sw} \cdot C_{in}}$$

- Will keep these, hard to get better without upsizing to 0805 or 1210

Previous Output Cap Selected did not provide DC bias derate curves -> switching to Murata Caps since they have published DC-bias curves.

GRM187R61A226ME15D (22µF output caps)

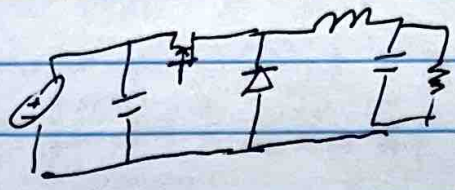


- Getting about 5.723*2 = 11.45µF at 5V output
- This now estimates a 9.7 mV pp Vout ripple
  - 0.2% Voltage ripple, well within parameters

(Using these at worst-case Vin = 15V, D = 33%)

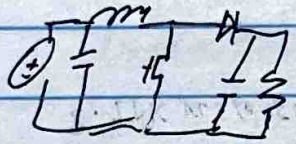
$$\Delta I_L = \frac{(V_{\text{IN}} - V_{\text{OUT}}) \cdot D}{f_s \cdot L} \qquad \Delta V_{\text{cap}} = \frac{\Delta I_L}{8 \cdot C_o \cdot f_s}$$

Next page: written notes during selection



BUCK CONVERTER

FLIP IT



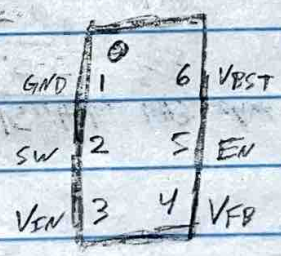
BOOST CONVERTER

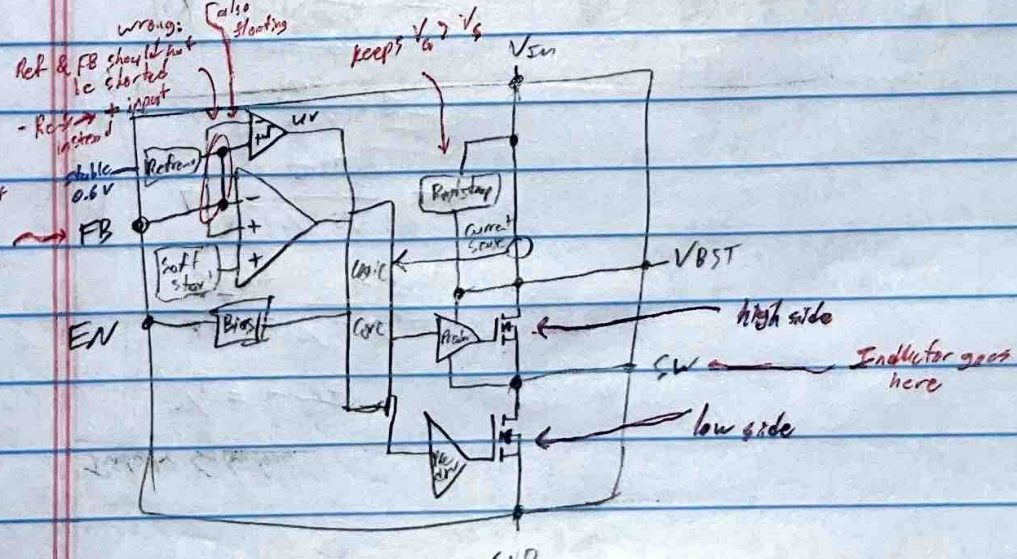
$V_{out} = \left(\frac{1}{1-D}\right) V_{in}$

$I_{out} = I_L (1-D)$

- period T / switching freq f_s affect the ~~efficiency~~ ripple
  - bigger f_s → less ripple → more switching losses
  - smaller f_s → more ripple → less switching losses

TPS30208



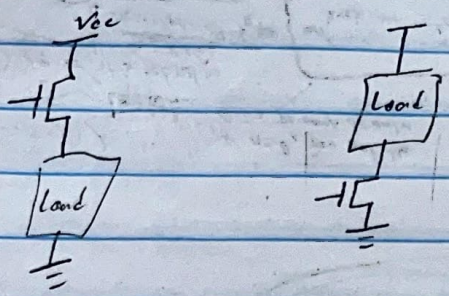


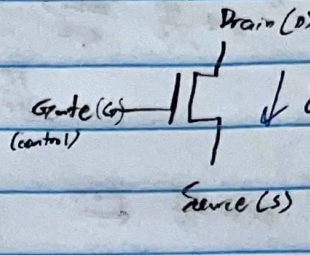
Logic Core

- commands gate drivers based off feedback and current sense
- dead time between switch transitions

has on R_{DS} like a R when on (~mΩ)

High Side · **Low Side**



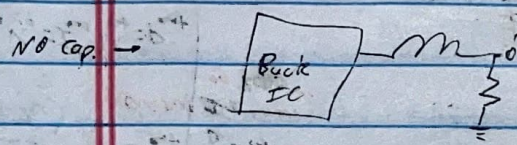


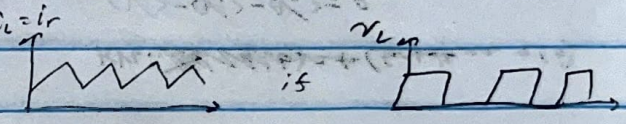
$V_{GS} > V_{GS,th}$ → turns on

NMOS

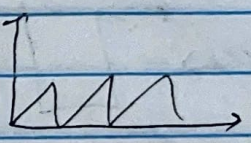
- μ_n electron mobility > μ_p hole mobility
  - ↳ NMOS R_{DS} < PMOS R_{DS}
    - → less power losses

Understanding Current & Voltage Ripple — Role of L and C

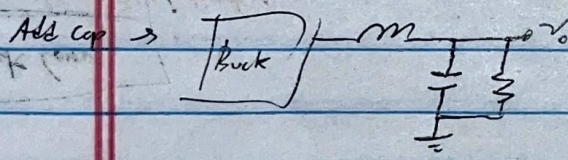




$v_o = i_r R$ →



BAD



now, $i_L = i_c + i_R$

$= i_c + \frac{v_o}{R}$

$i_L = C\frac{dv_o}{dt} + \frac{v_o}{R}$

ok test w/ $v_o = 5$ V, $R = 2.5\,\Omega$

$i_L = i_c + 2$

$i_c = i_L - 2$

$\frac{dv_o}{dt} = \frac{i_L - i_r}{C}$

↑ voltage ripple as i_L changes

and i_L could be idk 2A avg, 1A peak-peak

i_c can be + or −, pulling or pushing current back to maintain constant i_r??

- want $\frac{dv_o}{dt} \to 0$

Inductor Selection

$$V_L = L\frac{di}{dt}$$

$$\frac{\Delta i}{\Delta t_{on}} = \frac{V_{in} - V_{out}}{L} \quad \text{(when on)}$$

$$L = \frac{(V_{in} - V_{out})}{\Delta i_L}\Delta t_{on}$$

$$= \frac{V_{in} - V_o}{\Delta i_L} \cdot DT$$

$$\boxed{L = \frac{V_{in} V_o}{\Delta i} \cdot \frac{D}{f}} \qquad \left(L \propto \frac{1}{f}\right)$$

↖ suggested L

Datasheet uses a 15 μH L, let's check this

$$\Delta i = \frac{13.5 - 5}{15 \cdot 10^{-6}} \cdot \frac{5/13.5}{500 \cdot 10^3} \qquad \left(D = \frac{V_{out}}{V_{in}}\right)$$

$\Delta I_L \approx$ (0.420 A) peak-to-peak current ripple @ 13.5 V input @ 500 kHz

$\frac{0.42}{2} = 21\%$ of I_{out}

↳ peak current: $I_{peak} = 2 + \frac{0.42}{2} = \boxed{2.21\ A}$

15 μH Good.

↳ Two options

1. Eaton HCM1A0805V3-150-R, $I_{sat} = 4.6\ A$, 52 mΩ DCR, (8×8 mm)
2. TDK VLS6045EX-150M, $I_{sat} = 3.1\ A$, 75 mΩ DCR, (6×6 mm)

(both SMD)

Capacitor Selection(s)

$$\Delta V_{out} = \frac{\Delta I_L}{8 C_{out} f_s} + (\Delta I_L \cdot ESR)$$

↗ not considering at the moment

$$C_{out} = \frac{\Delta I_L}{8 \Delta V_{out} f_s}$$

$$= \frac{0.42}{(8)(0.05)(500 \times 10^3)}$$

1.–5 % voltage ripple

$\Delta V_{out} = 0.01(5\,V) = 0.05\,V$

$\Delta V_{out} = 0.05(5\,V) = 0.25\,V$

$C_{out} = 2.1\,\mu F$ min C for ~1% voltage ripple (theoretical)

BUT manufacturer suggests 22μF||22μF||100nF

↖ decouples high-f noise ($Z_c = \frac{1}{\omega C}$)

Why?

↳ Higher C accounts for DC-bias derating

ESR and ESL lower (in parallel)

Transients

Feedback Rs

need 0.6V

$$V_{FB} = V_{out} \frac{R_1}{R_1 + R_2}$$

$$\Rightarrow \frac{R_1}{R_1+R_2} = \frac{0.6}{5} = 0.12$$

$$\frac{3}{3+22} = 0.12$$

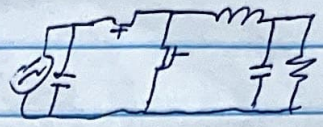
But $I = \frac{5}{3} = 1.67\,A$

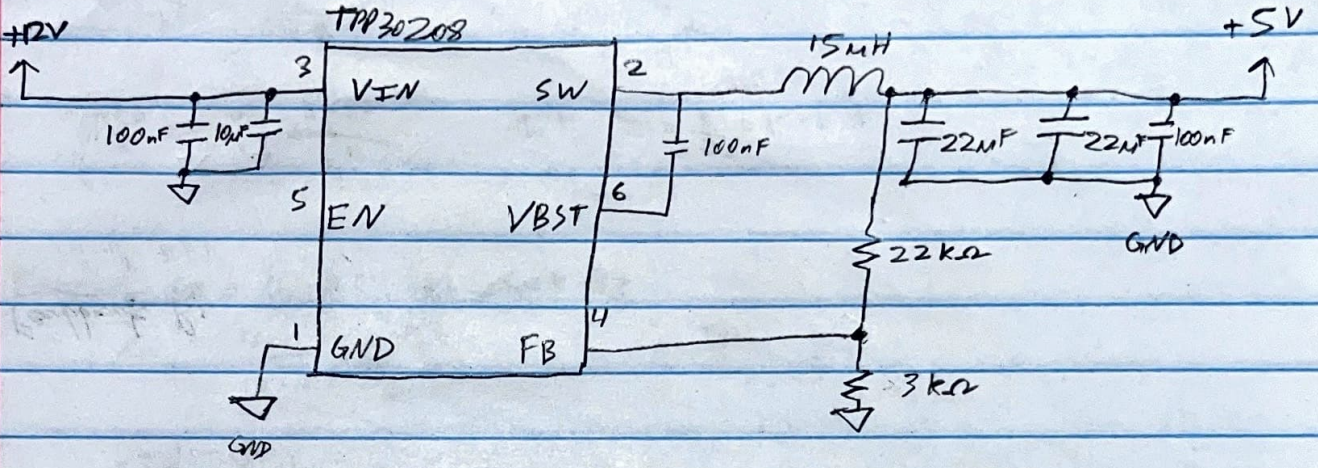
↓

$\frac{5}{3k} = 1.7\,mA$ ✓

3K and 22K

Design Circuit





Activity #3: Schematic and Layout Design

Here is where you start designing the schematic! Create a branch in the onboarding repo, and use the template KiCad project to make your buck converter. Be sure to follow the LV standards guide.

Please attend one of the KiCad workshops, where the electrical leads will teach you more about the software and what we expect.

Once you feel that your schematic and or layout is done, make a commit on the fs-electrical-onboarding repository, and push it onto github. You should make a commit no matter what, even if you did not finish completely. If you do not, you will not be considered for Power Electronics.